

Tri-Modal Sequence Model (TSM) - Complete Explanation

1. Core Idea

TSM combines Transformer (attention), Mamba (state memory), and Hyena (pattern filtering) into a unified system to overcome limitations of individual architectures.

2. Problem with Transformers

Transformers rely on attention which has quadratic complexity and no persistent memory. They reprocess tokens repeatedly and forget beyond context window.

3. Mamba Contribution

Mamba introduces a state-space model that maintains evolving memory, avoiding recomputation and enabling linear complexity processing.

4. Hyena Contribution

Hyena replaces attention with convolutional filters to capture long-range patterns efficiently without pairwise comparisons.

5. Why Combine Them

Each model solves a different problem: Transformer (precision), Mamba (memory), Hyena (efficiency). Combining them gives a multi-perspective understanding.

6. Representation Mismatch Problem

Each model outputs vectors with different meanings, scales, and structures. Direct combination leads to meaningless results.

7. Projection Solution

Projection maps outputs into a shared space using learned matrices, aligning dimensions and meanings for proper fusion.

8. Fusion Mechanism

Adaptive gating dynamically assigns importance to each model based on input context, enabling intelligent combination.

9. Core Challenges

Representation mismatch, fusion instability, training difficulty, module dominance, compute cost, and optimization complexity.

10. Advanced Challenges

Credit assignment, redundancy, temporal misalignment, scaling issues, evaluation difficulty, and implementation complexity.

11. Key Insight

The goal is not combining outputs, but aligning different types of intelligence into a unified representation.

12. Final Vision

Future AI systems will be hybrid architectures combining reasoning, memory, and pattern recognition into one system.